KAIZEN MATHS

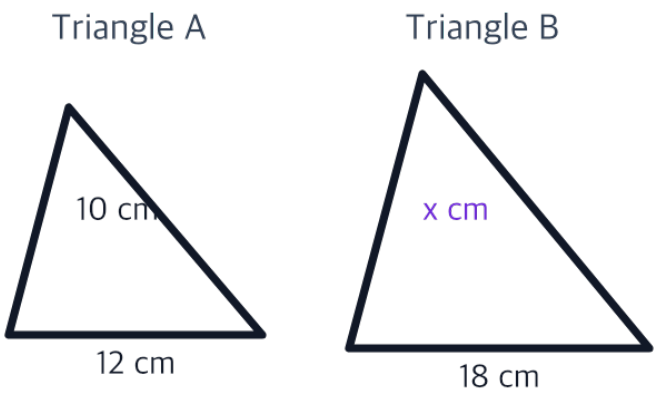
Lesson starter

Scale Drawings and Similar Shapes

Similar Triangles

Name _____

SA1



The triangles are similar. Find x.

GEO-9EE4-L3 · SA

Show your working

KAIZEN MATHS

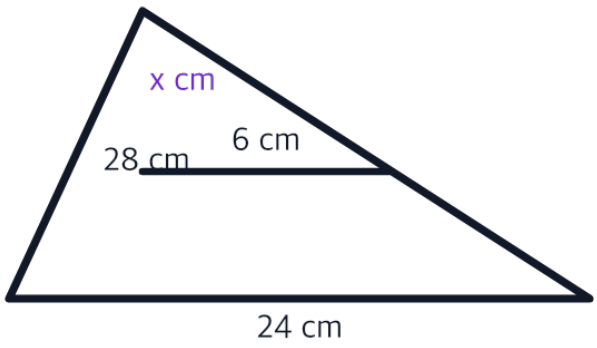
Lesson starter

Scale Drawings and Similar Shapes

Similar Triangles

Name _____

SB1



The line inside the triangle is parallel to the base, so the small triangle is similar to the whole triangle. Find x.

GEO-9EE4-L3 · SB

Show your working

KAIZEN MATHS

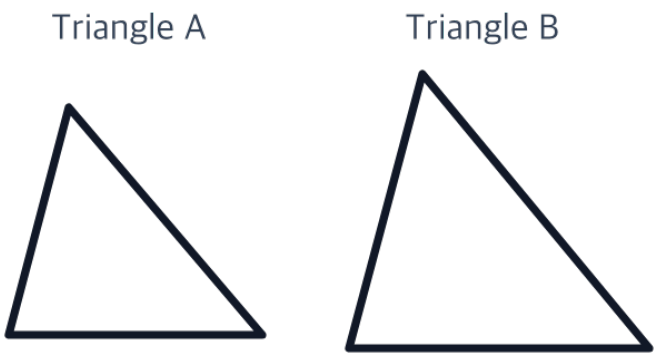
Lesson starter

Scale Drawings and Similar Shapes

Similar Triangles

Name _____

SC1



Two similar triangles have scale factor 2:1 from Triangle A to Triangle B. Triangle A has perimeter 33 cm. Find the perimeter of Triangle B.

GEO-9EE4-L3 · SC

Show your working

KAIZEN MATHS

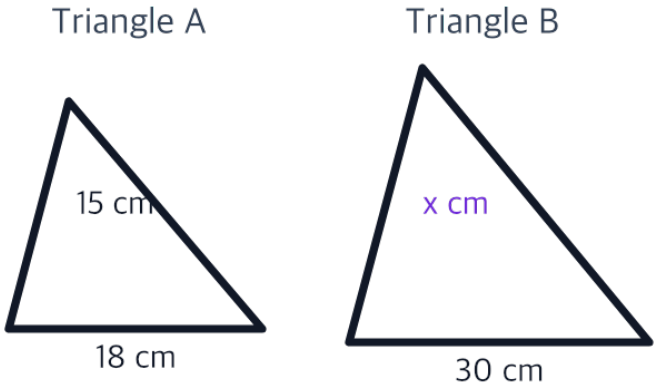
Lesson starter

Scale Drawings and Similar Shapes

Similar Triangles

Name _____

SD1



The triangles are similar. Find x.

GEO-9EE4-L3 · SD

Show your working

KAIZEN MATHS

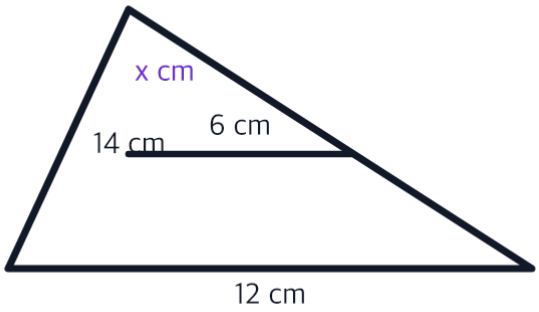
Exit ticket

Scale Drawings and Similar Shapes

Similar Triangles

Name _____

EA1



The line inside the triangle is parallel to the base, so the small triangle is similar to the whole triangle. Find x.

Show your working

KAIZEN MATHS

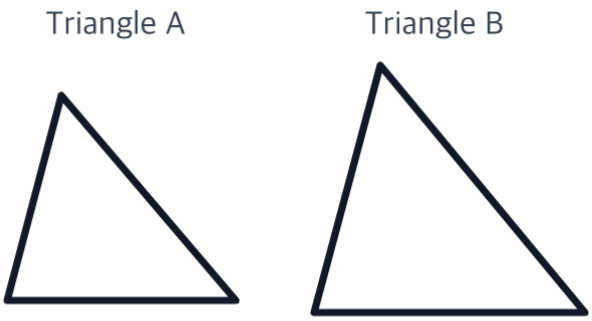
Exit ticket

Scale Drawings and Similar Shapes

Similar Triangles

Name _____

EB1



Two similar triangles have scale factor 3:1 from Triangle A to Triangle B. Triangle A has perimeter 24 cm. Find the perimeter of Triangle B.

Show your working

KAIZEN MATHS

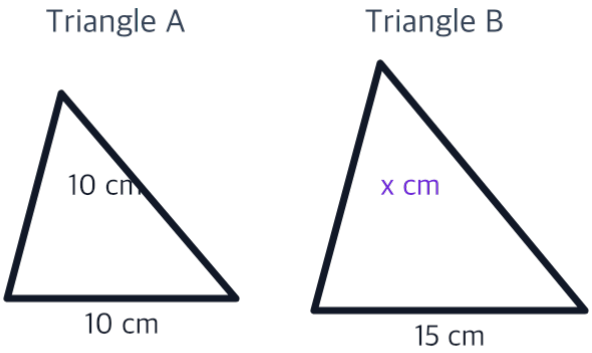
Exit ticket

Scale Drawings and Similar Shapes

Similar Triangles

Name _____

EC1



The triangles are similar. Find x.

Show your working

KAIZEN MATHS

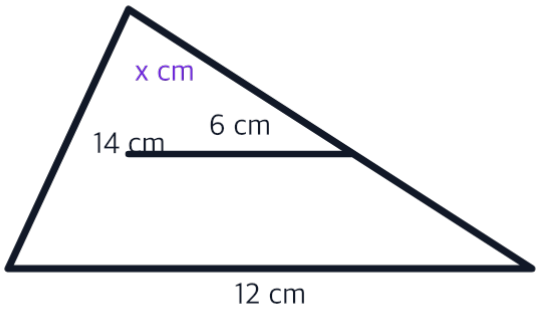
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Scale Drawings and Similar Shapes

Similar Triangles

Name _____

EA1



The line inside the triangle is parallel to the base, so the small triangle is similar to the whole triangle. Find x.

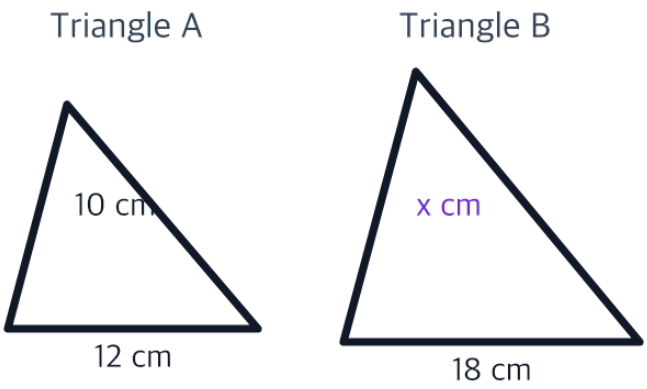
Show your working

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Triangles

GEO-9EE4-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1



The triangles are similar. Find x.

Answer:

15 cm

Working:

Identify corresponding sides before forming the ratio.

Scale factor from Triangle A to Triangle B = 18 ÷ 12 = 1.5.

x = corresponding side in Triangle A × scale factor.

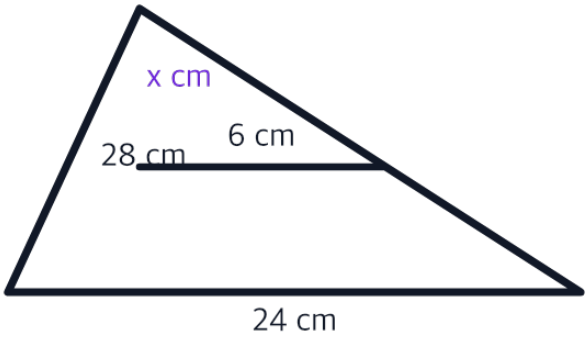
$x = 10 \times 1.5 = 15$ cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Triangles

GEO-9EE4-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SB1



The line inside the triangle is parallel to the base, so the small triangle is similar to the whole triangle. Find x.

Answer:

7 cm

Working:

The parallel line creates two similar triangles.

Scale factor from the small triangle to the whole triangle = whole base ÷ small base.

Scale factor = 24 ÷ 6 = 4.

Small side x = whole corresponding side ÷ 4.

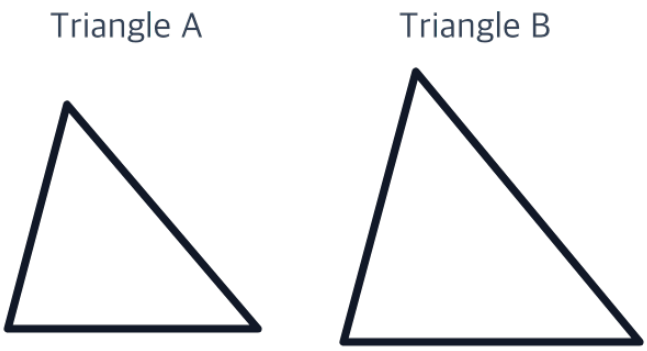
$x = 28 \div 4 = 7$ cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Triangles

GEO-9EE4-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SC1



Two similar triangles have scale factor 2:1 from Triangle A to Triangle B. Triangle A has perimeter 33 cm. Find the perimeter of Triangle B.

Answer:

66 cm

Working:

Perimeter uses the same scale factor as length.

Scale factor from A to B = 6 ÷ 3 = 2.

Perimeter of B = perimeter of A × scale factor.

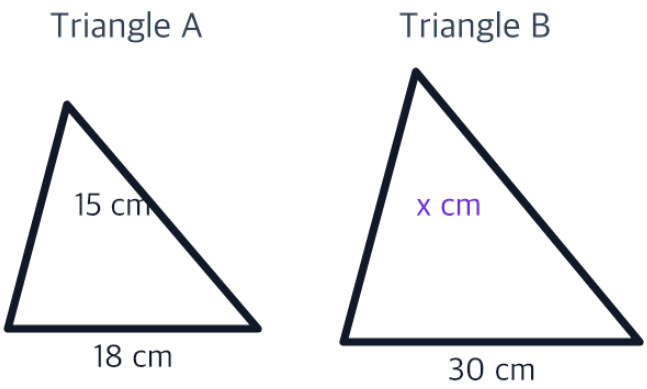
Perimeter of B = 33 × 6 ÷ 3 = 66 cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Triangles

GEO-9EE4-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SD1



The triangles are similar. Find x.

Answer:

25 cm

Working:

Identify corresponding sides before forming the ratio.

Scale factor from Triangle A to Triangle B = 30 ÷ 18 = 1.67.

x = corresponding side in Triangle A × scale factor.

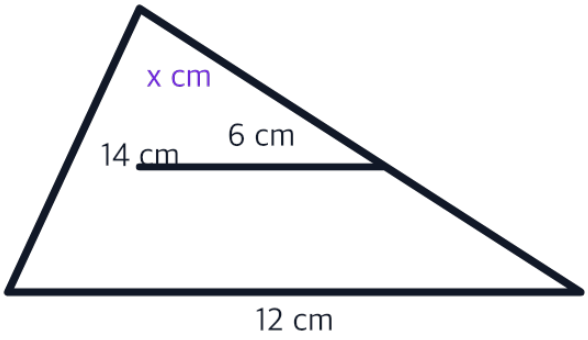
$x = 15 \times 1.67 = 25$ cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Triangles

GEO-9EE4-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EA1



The line inside the triangle is parallel to the base, so the small triangle is similar to the whole triangle. Find x.

Answer:

7 cm

Working:

The parallel line creates two similar triangles.

Scale factor from the small triangle to the whole triangle = whole base ÷ small base.

Scale factor = 12 ÷ 6 = 2.

Small side x = whole corresponding side ÷ 2.

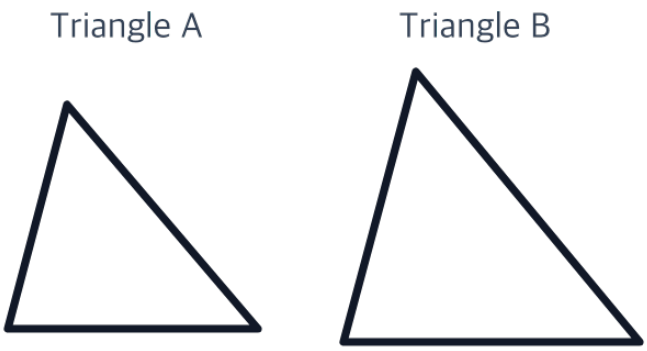
$x = 14 \div 2 = 7$ cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Triangles

GEO-9EE4-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EB1



Two similar triangles have scale factor 3:1 from Triangle A to Triangle B. Triangle A has perimeter 24 cm. Find the perimeter of Triangle B.

Answer:

72 cm

Working:

Perimeter uses the same scale factor as length.

Scale factor from A to B = 3 ÷ 1 = 3.

Perimeter of B = perimeter of A × scale factor.

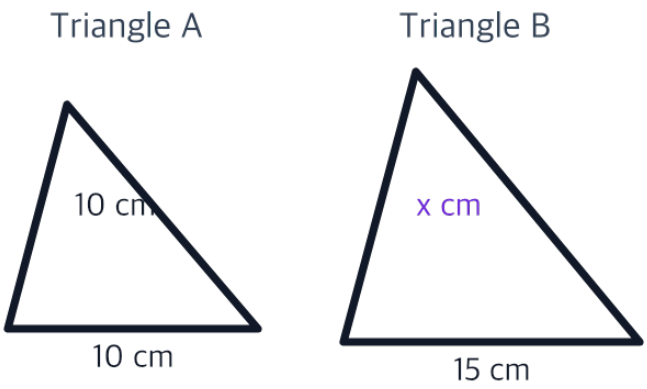
Perimeter of B = 24 × 3 ÷ 1 = 72 cm.

Teacher answers & working

Scale Drawings and Similar Shapes · Similar Triangles

GEO-9EE4-L3 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

EC1



The triangles are similar. Find x.

Answer:

15 cm

Working:

Identify corresponding sides before forming the ratio.

Scale factor from Triangle A to Triangle B = 15 ÷ 10 = 1.5.

x = corresponding side in Triangle A × scale factor.

$x = 10 \times 1.5 = 15$ cm.